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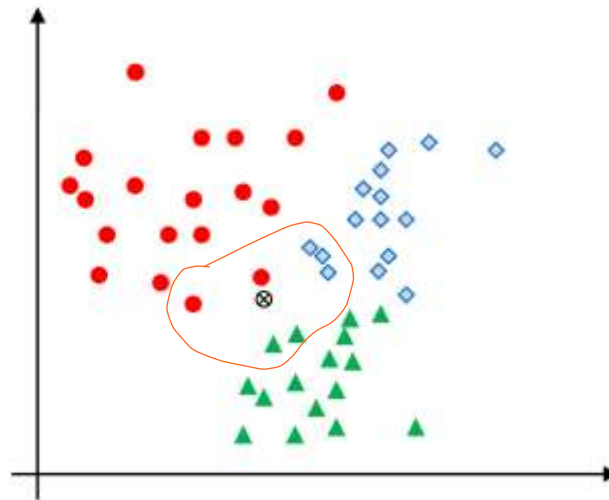
University ID Number

Signature

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- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
 - The number of pages is 7. Additional papers will not be considered.
 - Use of books, lecture notes or any other material is forbidden. Electronics (smartphones, calculators, etc.) are forbidden.
 - Clarity, order and precision are strongly considered for the final evaluation.
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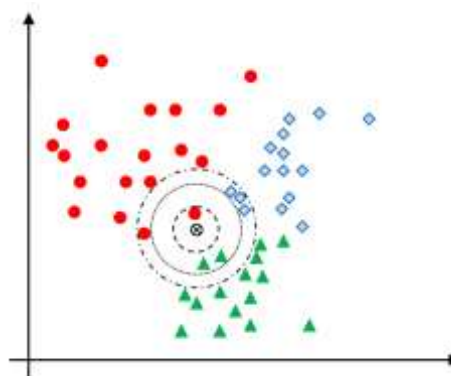
1. [k-nearest neighbors]

Given the data distributions in the figure below:



- 1.1 Say how the new point $\otimes$ will be classified by three k-NN models, with $k = 1$, $k = 3$ and $k = 7$, respectively. Use the Euclidean distance for all the models.
- 1.2 What would be a reasonable choice for k in the example?
- 1.3 Explain the main differences between the perceptron and the k-NN models for classification.

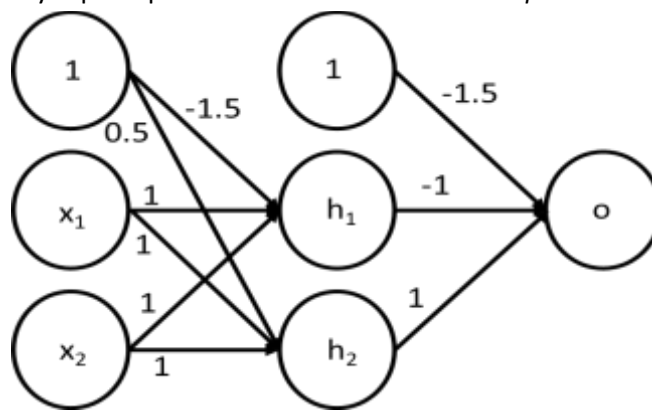
- 1.1 Given the prediction rules of the k-NN approach explained in Chapter 9 of the Lecture Notes, the new point will be assigned: the “red-circle class” if $k = 1$ (dashed neighborhood in the figure below), the “green-triangle class” if $k = 3$ (dotted neighborhood in the figure below) and the “blue-diamond class” if $k = 7$ (dash-dotted neighborhood in the figure below).



- 1.2 As indicated in Chapter 9 of the Lecture Notes, a good choice for k is $\sqrt{N}$, so here $k = 7$ and $k = 8$ may be reasonable values (notice that $N = 50$).
- 1.3 See Chapters 3 and 9 of the Lecture Notes.

2. [Neural networks]

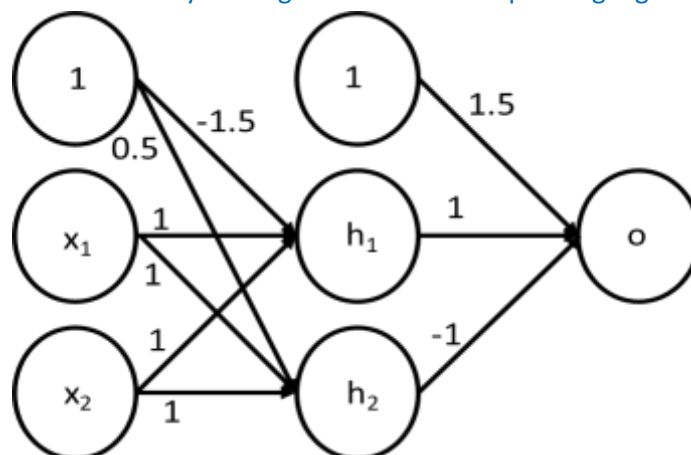
The figure below shows a multilayer perceptron that constructs the *XOR operator*.



- 2.1 How would you modify it to construct the *binary equivalence function* (i.e. such that the output is +1 only if both inputs are either -1 or 1)?
- 2.2 Describe the architecture of a general feed-forward neural network and explain why it is generally a more powerful modeling tool with respect to the perceptron and the logistic function in classification problems.

2.1 In the figure above, h_1 denotes the AND operator and h_2 denotes an operator that returns -1 only when both the inputs are negative. The output layer describes the AND between h_2 and the negation of h_1 , which is +1 only when one of the two inputs is +1 and the other is -1 (the so-called XOR logical function).

To obtain that required function, which is the negation of the XOR operator, it is sufficient to switch signs on all the output weights (this could also be verified by looking at the final corresponding logical table).



- 2.2 See Chapters 3 (perceptron), 6 (logistic regression) and 10 (neural networks) of the Lecture Notes.

3. [Kalman filtering]

Consider a constant random variable x , with zero mean and unitary variance. At each time instant t , a new noisy measure $y(t)$ of the variable x is available, according to the equation

$$y(t) = x + v(t),$$

where $v(t)$ is a white noise with zero mean and unitary variance, uncorrelated with x . Assume that an estimate $\hat{x}(1)$ of the initial value of x is given, with known corresponding variance $P(1) = \text{var}(x - \hat{x}(1))$.

3.1 At time $t = 1$, a measurement $y(1)$ becomes available. Compute $\hat{x}(2|1)$ and its variance $P(2) = \text{var}(x - \hat{x}(2|1))$.

3.2 At time $t = 2$, a new measurement $y(2)$ becomes available. Compute $\hat{x}(3|2)$ and its variance $P(3) = \text{var}(x - \hat{x}(3|2))$.

3.3 Analyze the convergence properties of $P(t)$.

3.4 Draw the block scheme of the predictor of x .

3.1 Since x is constant, the state dynamics is simply given by $x(t+1) = x(t)$. The system matrices are then $F = 1, H = 1, V_1 = 0, V_{12} = 0, V_2 = 1$, while the DRE becomes

$$P(t+1) = P(t) - P(t)(P(t) + 1)^{-1}P(t) = \frac{P(t)}{P(t) + 1}$$

The corresponding Kalman predictor then reads as

$$\begin{cases} \hat{x}(t+1|t) = F\hat{x}(t|t-1) + K(t)(y(t) - \hat{y}(t|t-1)) \\ \hat{y}(t|t-1) = H\hat{x}(t|t-1) \end{cases}$$

where the gain is given by

$$K(t) = \frac{P(t)}{P(t) + 1}$$

It follows that, using the available information,

$$\begin{aligned} \hat{x}(2|1) &= \hat{x}(1) + \frac{P(1)}{P(1) + 1}(y(1) - \hat{x}(1)) \\ P(2) &= \frac{P(1)}{P(1) + 1} \end{aligned}$$

3.2 Similarly to the previous point:

$$\begin{aligned} \hat{x}(3|2) &= \hat{x}(2|1) + \frac{P(2)}{P(2) + 1}(y(2) - \hat{x}(2|1)) \\ P(3) &= \frac{P(2)}{P(2) + 1} \end{aligned}$$

3.3 Since

$$P(t+1) = \frac{P(t)}{P(t) + 1} < P(t), \forall t$$

the DRE converges to the solution of the ARE, $\bar{P} = 0$, for any initial admissible condition. This can also be checked via the graphical method.

3.4 See Chapter 15 of the Lecture Notes.

4. [Learning with time series]

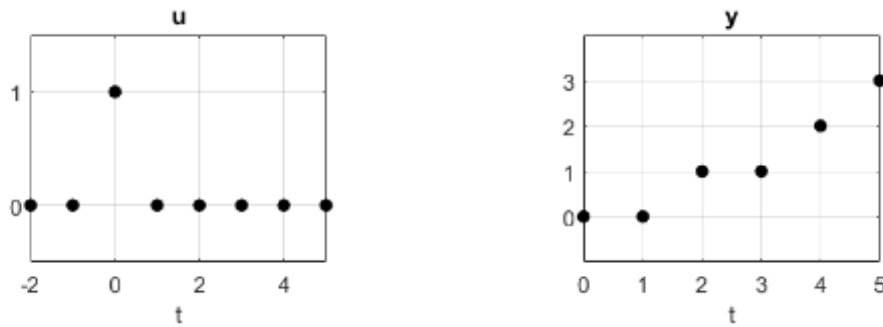
Describe a possible sample estimator for the covariance function of a stationary stochastic process. Is it a correct estimator? Why? What are the main limitations of such an estimator due to the finiteness of the dataset?

Solutions:

See Chapter 14 of the Lecture Notes.

5. [Subspace methods]

Consider the system that given the input u (left figure) generates the output y (right figure).



Starting from this dataset: 1) Identify the system order. 2) Compute the system transfer function. 3) Check if the system is asymptotically stable.

Identify the system order

$$H_1 = [0] \rightarrow \text{rank}(H_1) = 0$$

$$H_2 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \rightarrow \text{rank}(H_2) = 2$$

$$H_3 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} \rightarrow H_3(3,:) = H_3(1,:) + H_3(2,:) \rightarrow \text{rank}(H_3) = 2 \rightarrow n = 2$$

Compute the system transfer function

$$H_3 \text{ factorization} \rightarrow \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}}_{O_3} \underbrace{\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}}_{R_3} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\text{Matrix extraction} \rightarrow \begin{aligned} F &= O_3(1:n,:)^{-1} O_3(2:n+1,:) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} & G &= R_3(:,1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ H &= O_3(1,:) = [1 \quad 0] & D &= \omega(0) = 0 \end{aligned}$$

$$\text{Transfer function} \rightarrow W(z) = H(zI - F)^{-1}G + D = \dots = \frac{1}{z^2 - z - 1}$$

Check if the system is asymptotically stable

The impulse response diverges then the system is not asymptotically stable. For a formal check we must verify if all the poles of the system transfer function are inside the unitary circle.

$$z^2 - z - 1 = 0 \rightarrow z_{1,2} = \frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

$$z = \frac{1}{2} + \frac{\sqrt{5}}{2} > 1 \rightarrow \text{the system is not asymptotically stable}$$

6. [Analysis and prediction of ARMAX processes]

Consider the process: $y(t) = \frac{(z - \sqrt{3} + z^{-1})(1 + 5z^{-1})}{(1 + \frac{1}{5}z^{-1})(1 - \frac{1}{2}z^{-1})} e(t - 1)$ $e(t) \sim WN\left(\frac{1}{5}, \frac{1}{25}\right)$

- 1) Discuss the stationarity of process $y(t)$. 2) Compute the mean value and the variance of process $y(t)$. 3) Compute and draw the spectrum of process $y(t)$

Discuss the stationarity of process $y(t)$

The process is equivalent to $y(t) = \frac{(1 - \sqrt{3}z^{-1} + z^{-2})}{(1 - \frac{1}{2}z^{-1})} \eta(t)$ where $\eta(t) \sim WN(1, 1)$

$y(t)$ is a stationary stochastic process since $W(z)$ is asymptotically stable (pole $z = \frac{1}{2}$) and $\eta(t)$ is a stationary stochastic process (it is a WN).

Compute the mean value and the variance of process $y(t)$

$$y(t) = \underbrace{\frac{(1 - \sqrt{3}z^{-1} + z^{-2})}{(1 - \frac{1}{2}z^{-1})} \tilde{\eta}(t)}_{\tilde{y}(t)} + \underbrace{2(2 - \sqrt{3})}_{m_y}$$

$$\tilde{y}(t) = \frac{1}{2} \tilde{y}(t-1) + \tilde{\eta}(t) - \sqrt{3} \tilde{\eta}(t-1) + \tilde{\eta}(t-2)$$

$$\begin{aligned} \gamma_y(0) &= E[\tilde{y}(t)^2] = E\left[\left(\frac{1}{2} \tilde{y}(t-1) + \tilde{\eta}(t) - \sqrt{3} \tilde{\eta}(t-1) + \tilde{\eta}(t-2)\right)^2\right] = \\ &= \frac{1}{4} \gamma_y(0) + \underbrace{E[\tilde{\eta}(t)^2]}_1 + 3 \underbrace{E[\tilde{\eta}(t)^2]}_1 + \underbrace{E[\tilde{\eta}(t)^2]}_1 - \sqrt{3} \underbrace{E[\tilde{y}(t-1) \tilde{\eta}(t-1)]}_1 + \underbrace{E[\tilde{y}(t-1) \tilde{\eta}(t-2)]}_{\frac{1}{2} - \sqrt{3}} \end{aligned}$$

$$\gamma_y(0) = \frac{\frac{11}{2} - 2\sqrt{3}}{\frac{3}{4}} \cong 2.71$$

Compute and draw the spectrum of process $y(t)$

$$\Gamma_y(\omega) = W(e^{-j\omega}) W(e^{j\omega}) \Gamma_\eta(\omega)$$

$$\Gamma_y(\omega) = \frac{(1 - \sqrt{3}e^{-j\omega} + e^{-2j\omega})(1 - \sqrt{3}e^{j\omega} + e^{2j\omega})}{(1 - \frac{1}{2}e^{-j\omega})(1 - \frac{1}{2}e^{j\omega})} =$$

$$\Gamma_y(\omega) = \frac{5 - 2\sqrt{3}(e^{j\omega} + e^{-j\omega}) + (e^{2j\omega} + e^{-2j\omega})}{\frac{5}{4} - \frac{1}{2}(e^{j\omega} + e^{-j\omega})} =$$

$$\Gamma_y(\omega) = \frac{5 - 4\sqrt{3} \cos(\omega) + 2 \cos(2\omega)}{\frac{5}{4} - \cos(\omega)}$$

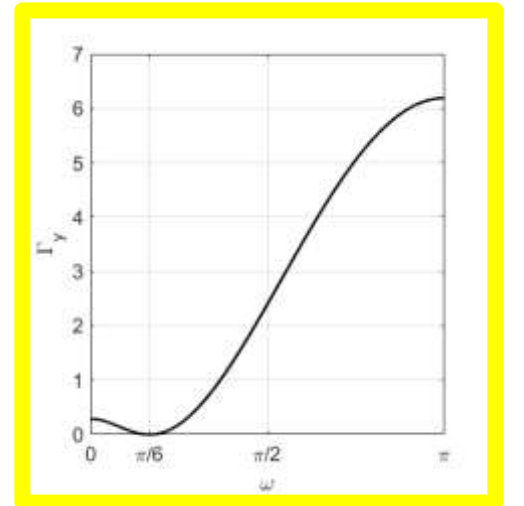
It is necessary to evaluate $\Gamma_y(\omega)$ in $\omega \in \{0, \frac{\pi}{6}, \frac{\pi}{2}, \pi\}$

$$\Gamma_y(0) = \frac{5 - 4\sqrt{3} + 2}{\frac{5}{4} - 1} \cong 0.287$$

$$\Gamma_y\left(\frac{\pi}{6}\right) = \frac{5 - 4\sqrt{3} \frac{\sqrt{3}}{2} + 1}{\frac{5}{4} - \frac{\sqrt{3}}{2}} = 0$$

$$\Gamma_y\left(\frac{\pi}{2}\right) = \frac{12}{5} = 2.4$$

$$\Gamma_y(\pi) = \frac{7 + 4\sqrt{3}}{\frac{5}{4} + 1} \cong 6.19$$



7. [Identification of ARMAX processes]

Consider a model belonging to the model class

$$\mathcal{M}(\theta): y(t) = ay(t-1) + u(t-1) + b + \xi(t) \quad \xi(t) \sim WN(0, \lambda_\xi^2), \text{ where } \theta = \begin{bmatrix} a \\ b \end{bmatrix} \text{ is the parameter vector.}$$

Suppose that the following measurements are available

t	0	1	2	3
u	0	1	1	5/8
y	0	1	3/2	5/4

Find the vector θ^* which minimizes $J(\theta) = \frac{1}{N} \sum_{t=1}^N (y(t) - \hat{y}(t|t-1))^2$, and compute the predictor $\hat{y}(4|3)$

$$\hat{y}(t|t-1) = ay(t-1) + u(t-1) + b$$

$$\hat{y}(1|0) = ay(0) + u(0) + b = b$$

$$\hat{y}(2|1) = ay(1) + u(1) + b = a + b + 1$$

$$\hat{y}(3|2) = ay(2) + u(2) + b = \frac{3}{2}a + b + 1$$

t	0	1	2	3
$y(t) - \hat{y}(t t-1)$	/	$1 - b$	$\frac{1}{2} - a - b$	$\frac{1}{4} - \frac{3}{2}a - b$

$$J(\theta) = \frac{1}{3} \left[(1-b)^2 + \left(\frac{1}{2} - a - b \right)^2 + \left(\frac{1}{4} - \frac{3}{2}a - b \right)^2 \right]$$

$$J(\theta) = \frac{1}{3} \left[1 + \frac{1}{4} + \frac{1}{16} + \frac{13}{4}a^2 + 3b^2 + 5ab - \frac{7}{4}a - \frac{7}{2}b \right]$$

$$\frac{\partial J}{\partial a} = 0 \rightarrow \frac{1}{3} \left[\frac{13}{2}a + 5b - \frac{7}{4} \right] = 0$$

$$\frac{\partial J}{\partial b} = 0 \rightarrow \frac{1}{3} \left[6b + 5a - \frac{7}{2} \right] = 0$$

$$\rightarrow \begin{cases} \frac{13}{2}a + 5b - \frac{7}{4} = 0 \\ 6b + 5a - \frac{7}{2} = 0 \end{cases}$$

$$\rightarrow \begin{cases} a = -\frac{1}{2} \\ b = 1 \end{cases}$$

$$\hat{y}(4|3) = 1$$